# Tejaswini Pedapati

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## Summary

Senior Research Software Engineer who spent the last decade spearheading efforts in **RL based alignment (Post-Training), LLM Safety, model pruning, distillation, AutoML and** with strong benchmark performance and production impact. Published in top-tier venues including **ACL, NeurIPS, ICML, ICLR and Nature Communications**. Was instrumental in multiple client engagements.

## Experience

**Senior Research Software Engineer, IBM Research** Feb 2014 – Present

**Post-Training and LLM Safety**

- Developed the **Granite Guardian** guardrail models by post-training the Granite family to detect harms such as toxicity, bias, jailbreaking and hallucinations.
- These models topped the **GuardBench** leaderboard and ranked third on **LLM Aggrefact**, earning an **IBM Outstanding Technical Achievement Award**.
- Led the efforts to prune **Granite Guardian 3.2 8B** model to **5B** with minimal drop in performance (<2%) across all the benchmarks and devised a novel confidence based distillation algorithm to recover the performance.
- Improved the performance of **Qwen3 8B** and **DS-R1-Distill-LLaMA 8B** models on safety benchmarks by 12% through post-training with safety data using **GRPO** and using **granite 3.2 5B** as a reward model.
- Contributed novel contrastive explainability algorithms for deep learning and machine learning classifiers to IBM's explainability toolkit and received an **IBM Outstanding Technical Achievement Award**.

**Efficient LLMs**

- **Efficient inference:** Led the development of novel algorithms for model, activation and KV cache pruning that reduced inference latency up to 9% and memory usage up to 17%.
- Designed adaptive inference budgeting methods to estimate reasoning ("thinking") budgets for LLMs, improving cost-efficiency.
- **Efficient model training:** Devised algorithms for training data subset selection, long context understanding and knowledge distillation that accelerated the training speed by up to 10 times.
- Explored cascading diverse domains into various stages, to minimize interference during post-training a model using RL.

**AutoML**

- Architected a modular **Neural Architecture Search (NAS)** framework that supported multiple search algorithms and resilient state recovery, which was later released as an IBM product for image and text modalities.
- Led two client engagements to train bespoke models for legal domain (text) and electronics domain (images).
- Improved search efficiency by 100 times using a meta-learning-based early stopping mechanism to filter out low-performing candidates.
- Developed a meta-learning algorithm to automate feature engineering for tabular datasets.
- All these contributions were honored with an **IBM Outstanding Technical Achievement Award**.

**Software Developer Intern, Yahoo Inc.** Jun 2013 – Aug 2013
**Software Developer, ThoughtWorks Technologies Ltd, India** Aug 2008 – Jun 2012

## Select Publications

- **Granite Guardian: Comprehensive LLM Safeguarding**, Inkit Padhi*, Manish Nagireddy*, Giandomenico Cornacchia*, Subhajit Chaudhury*, **Tejaswini Pedapati***, et al., NAACL Industry track, 2025
- **EPMAN: Episodic Memory Attention for Generalizing to Longer Contexts**, Subhajit Chaudhury*, Payel Das*, Sarathkrishna Swaminathan, Georgios Kollias, Elliot Nelson, Khushbu Pahwa, **Tejaswini Pedapati**, Igor Melnyk, Matthew Riemer, ACL 2025
- **Large Language Model Confidence Estimation via Black-Box Access**, **Tejaswini Pedapati**, Amit Dhurandhar, Soumya Ghosh, Soham Dan, Prasanna Sattigeri, TMLR 2025
- **Differentiable Prompt Learning for Vision Language Models**, Zhenhan Huang, **Tejaswini Pedapati**, Pin-Yu Chen, Jianxi Gao, IJCAI Intelligence 2025
- **From PEFT to DEFT: Parameter Efficient Finetuning for Reducing Activation Density in Transformers**, Bharat Runwal, **Tejaswini Pedapati**, Pin-Yu Chen, AAAI 2025
- **Network properties determine neural network performance**, Chunheng Jiang, Zhenhan Huang, **Tejaswini Pedapati**, Pin-Yu Chen, Yizhou Sun, Jianxi Gao, Nature Communications 2024
- **NeuroPrune: A Neuro-inspired Topological Sparse Training Algorithm for Large Language Models**, Amit Dhurandhar*, **Tejaswini Pedapati***, Ronny Luss*, Soham Dan, Aurelie Lozano, Payel Das, Georgios Kollias, ACL Findings 2024
- **CoFrNets: Interpretable Neural Architecture Inspired by Continued Fractions**, Isha Puri*, Amit Dhurandhar*, **Tejaswini Pedapati**, Karthikeyan Shanmugam, Dennis Wei, Kush R. Varshney, Neurips 2021
- **Learning Global Transparent Models Consistent with Local Contrastive Explanations**, **Tejaswini Pedapati**, Avinash Balakrishnan, Karthikeyan Shanmugam, Amit Dhurandhar, Neurips 2020
- **Learning to Rank Learning Curves**, Martin Wistuba and **Tejaswini Pedapati**, ICML 2020

## Select Talks

- Invited talk at Women in Data Science Conference 2025: "Challenges in Long Context LLMs".
- Invited talk at Open Data Science Conference East 2023: "Introduction to AutoML: Hyperparameter Optimization and Neural Architecture Search".
- Tutorial at AAAI 2021: "Efficient Neural Architecture Search".
- Talk at ICMR 2020: "Automation of Deep Learning-Theory and Practice".
- Talk at CVPR 2020: "Transfer Learning for Neural Architecture Search".

## Education

**M.S. in Computer Science (NLP & ML) | Columbia University** — Sep 2012 – Dec 2013
**B.Tech in Information Technology | PSG College Of Technology, India** — Aug 2004 – May 2008

## Technical Skills

**Core Expertise:** Alignment, Post-Training, RL, Model Pruning, Distillation, AutoML, Distributed Training
**Deep Learning Frameworks:** Pytorch, Deepspeed, Accelerate, FSDP, Ray, verl, TRL, VLLM